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Prediction of MIGS outcomes on second eyes using first eyes' response

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Abstract

To evaluate how well outcomes following cataract extraction and microinvasive glaucoma surgery in one eye predict outcomes in sequential second eye. Retrospective study of 78 patients who underwent cataract extraction and microinvasive glaucoma surgery in both eyes. Linear regressions using Pearson correlation coefficients were used to evaluate correlations in intraocular pressure and glaucoma medication change between eyes. Multivariable logistic regression models were used to evaluate the associations between first-eye variables and the likelihood of second-eye surgical success at 6 months. Surgical success was defined as meeting target intraocular pressure without additional medications compared to baseline or secondary surgical interventions. Baseline ocular characteristics were comparable between fellow eyes, with the majority having mild glaucoma. Intraocular pressure changes between fellow eyes at 6 months were modestly correlated between eyes ($R = 0.48$; $P < .001$). Changes in glaucoma medications were strongly correlated between eyes at all time points, and month 6 demonstrated the most significant correlation ($R = 0.80$; $P < .001$). First and second eye cohorts achieved 82% and 83% surgical success. Multivariate analysis for predictive factors of successful second eye surgery showed patients with successful first eye surgery at 6 months were significantly more likely to have successful second eye surgery (odds ratio, 20.67; $P < .001$). Reductions in intraocular pressure and glaucoma medications at 6 months following surgery in first eyes are correlated to second eye reductions. Successful surgical outcomes at 6 months following first eye surgeries are strongly associated with successful sequential second eye outcomes.

Abbreviations: CD = cup-to-disc, CI = confidence interval, HVF = Humphrey Visual Field, IOP = intraocular pressure, MD = mean deviation, ORs = odds ratios, SSIs = secondary surgical interventions.

Keywords: bilateral surgery, goniotomy, microinvasive glaucoma surgery, microstent, primary open angle glaucoma

1. Introduction

Glaucoma is one of the leading causes of irreversible vision loss.[1] In 2020, approximately 76 million individuals were affected by glaucoma.[2] By 2040, this estimate is expected to reach nearly 111.8 million individuals. Although glaucoma poses a significant threat to public health on a global scale, the underlying pathogenesis of glaucoma is not entirely understood. Therapeutic approaches to glaucoma concentrate on slowing the progression of optic nerve damage through intraocular pressure (IOP) control. Historically, the primary methods for IOP management included ocular hypotensive medications, laser trabeculoplasties, and traditional glaucoma filtering surgeries such as trabeculectomy and tube shunt. However, microinvasive glaucoma surgeries (MIGS) recently emerged as a promising alternative for IOP and ocular hypotensive medication reduction with a lower incidence of complications and side effects.[3] MIGS provides considerable IOP-lowering advantages while avoiding serious complications and detrimental damage to surrounding tissue structures.

The ophthalmic medical community has recently recognized MIGS for its potential to revolutionize glaucoma treatment and improve the quality of life for millions of people affected by this condition. A recent study using Intelligent Research in Sight (IRIS) Registry data reported that 75,358 eyes underwent concurrent microinvasive glaucoma surgeries (MIGS)

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The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

We observed that for patients requiring bilateral cataract extraction and microinvasive glaucoma surgeries, individuals with successful first-eye surgeries after 6 months are significantly more likely to have successful second-eye surgeries.

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and phacoemulsification procedures between 2013 and 2017.[4] Moreover, the number of MIGS procedures performed per year increased 3-fold from 2013 to 2017. With the estimated number of MIGS per year anticipated to continue to increase, the identification of predictive factors correlated with successful MIGS outcomes has the potential to aid clinicians in the selection of surgical interventions for patients with glaucoma as well as which patients are likely to benefit from MIGS.

Previous investigative studies have evaluated a variety of tools to predict outcomes of MIGS, including anterior segment optical coherence tomography angiography for trabecular-meshwork targeted MIGS and a netarsudil trial for micropulse transscleral laser therapy.[5,6] Other studies have used fellow eye analysis to contrast the efficacy of various glaucoma surgical approaches.[7,8] While these studies highlight the utility of intra-patient comparison for assessing the efficacy of different surgical techniques, there is an absence of literature analyzing whether first-eye MIGS outcomes can predict outcomes in the second eye.

In this study, we evaluate whether clinical outcomes following initial concurrent phacoemulsification and MIGS in patients' first eyes can predict outcomes in patients' second eyes. We also analyze the time points at which the correlation between first and second-eye MIGS results are the strongest and the potential demographic and clinical characteristics associated with favorable clinical outcomes. In addition, this analysis includes an evaluation of early postoperative time points that occur before second eye surgery for patients requiring bilateral phacoemulsification and MIGS.

2. Materials and methods

2.1. Ethics approval and patient consent

This study was reviewed and approved by the Johns Hopkins Institutional Review Board. Study practices were carried out in compliance with the tenets of the Declaration of Helsinki. Written informed consent from patients was not required.

2.2. Study population

A retrospective cohort study was conducted by evaluating the electronic medical records of glaucoma patients who underwent sequential phacoemulsification in combination with MIGS surgery (phaco-MIGS) in both eyes from February 2018 to April 2022. Patient records were extracted from a database of MIGS glaucoma surgeries performed at the Wilmer Eye Institute (Johns Hopkins University) from 2017 to 2023. All study participants within the database who received the same type of MIGS in both eyes were included. A total of 5 surgeons contributed to the procedures performed on study patients. Patients without the same type of MIGS in both eyes or missing 6-month follow-up visits were excluded. Four patients were excluded because of different MIGS in their eyes, and 41 patients were excluded because of unavailable 6-month follow-up data. Two patients were missing data for 3-month visits. Data for all other postoperative visits through 6 months were available for all patients.

2.3. Demographic and ocular characteristics

Baseline demographic variables included age, sex, and race/ethnicity as documented in the electronic health records. Baseline ocular variables comprised IOP, target IOP, type of glaucoma, glaucoma severity, the number and type of ocular hypotensive medications, prior glaucoma procedures, vertical cup-to-disc (C:D) ratio as determined by the optical coherence tomography, and Humphrey Visual Field (HVF) 24-2 mean deviation (MD). The type of Schlemm canal-based MIGS performed was also extracted from patient health records. MD values were used to determine the stage of glaucoma, with ≤−6 dB as mild, between −6.1 to −12 dB as moderate, and >12 dB as severe severity.[9] HVF data obtained earlier than a year from the surgery date was not documented. If HVF was not recorded, severity was determined by C:D ratio based on modified thresholds recommended by the Canadian Ophthalmological Society, with values under 0.65 documented as mild, <0.85 as moderate, and >0.9 as severe.[9] Of the first and second eyes, 9 and 11 eyes per cohort had unavailable HVF data. Target IOP was preoperatively established and recorded in the medical record by the treating surgeon based on the severity and stability of glaucoma and the baseline IOP. If data for target IOP were unavailable, mild, moderate, and severe glaucoma were assigned target IOP goal measurements of ≤18 mm Hg, ≤15 mm Hg, and ≤12 mm Hg, respectively.[10] Of first eye and second eyes, 9 and 10 eyes in each cohort, were missing target IOP.

2.4. Follow-up

The number and type of ocular hypotensive medications, IOP, adverse events, and any secondary surgical interventions (SSIs) were documented at all postoperative visits. The addition or discontinuation of ocular hypotensive medications was determined by each physician's clinical decision. Postoperative visits were typically conducted at 1 day, 1 week, 1 month, 3 months, and 6 months. Time intervals allotted for documenting postoperative visits were grouped by the following ranges: 1 day (1-day visit), 5 days to 15 days (1-week visit), 0.5 to 2 months (1-month visit), 2 to 4.5 months (3-months visit), and 4.5 to 7.5 months (6-months visit).

2.5. Study devices

Surgery was performed with topical anesthesia under intravenous sedation. A keratome of either 2.4 mm or 2.75 mm was used to make a clear corneal incision. Viscoelastic was injected through the incision to inflate and stabilize the anterior chamber. A Swan-Jacob gonioprism was placed on the cornea, and the patient's head was tilted to visualize the nasal angle. MIGS procedures, including Hydrus microstent implantation,[11] canaloplasty using the OMNI Surgical System,[12] bent ab-interno needle goniotomy,[13] Kahook Dual Blade goniotomy,[14] and iStent inject implantation,[15] were performed according to their instructions for use or previously described methods. If a patient received a combined canaloplasty using OMNI Surgical System with Hydrus Microstent, a 180-degree ab-interno canaloplasty was performed first. The microstent was implanted then through the initial goniotomy site.[16] Phacoemulsification with an intraocular lens was performed in the standard manner prior to or following MIGS.

2.6. Outcome measures

The primary outcome measure was the proportion of second eyes that achieved surgical success at 6 months compared to success in the first eye at postoperative time points. Surgical success was defined as meeting the target IOP without additional glaucoma medications from baseline or SSIs. For patients with baseline IOPs below their target IOPs, a reduction of at least one medication was required. Additional IOP-lowering surgeries or lasers were regarded as SSIs. Secondary outcome measures included correlations of change in IOP and the number of medications between the first and second eyes.

Covariates included patient demographic and baseline ocular characteristics. Demographic characteristics included race/ethnicity, sex, and age. Race/ethnicity was recorded based on categories reported in the electronic health records. Ocular covariates included glaucoma type, glaucoma severity, C:D ratio, MD values, number of baseline glaucoma medications, preoperative IOP, target IOP, surgeon, and type of MIGS surgery.

2.7. Statistical analysis

Correlations of changes in IOP and glaucoma medications between the study and fellow eyes were analyzed using paired t-tests, linear regression analyses, and Pearson correlations. Values of P equal to or <0.05 were considered statistically significant. Analysis of the association between the proportion of surgical success in the first eye at various postoperative time points and second eye success at 6 months was performed using the McNemar test.[17] We developed univariate and multivariable logistic regression models to calculate odds ratios (ORs) to evaluate the associations between first-eye variables and first-eye success at various time points and the likelihood of surgical success in the second eye at 6 months. Factors with P values <.10 in univariate analysis were included as covariates in the multivariable model. Success was defined as reaching the target IOP, no additional glaucoma medications compared to baseline, and no SSIs. Kaplan–Meier survival analysis and log-rank tests were used to evaluate success probabilities for first and second-eye phaco-MIGS. Summary statistics and regressions were generated using R 4.2.0 (R Foundation, Indianapolis, IN).

3. Results

3.1. Study characteristics

A total of 78 glaucoma patients were identified who underwent combined phacoemulsification and MIGS procedures in both eyes. Patient demographics and ocular characteristics are shown in Tables 1 and 2. A majority of the study population was female sex and of non-Hispanic white race/ethnicity. The mean age was 73.1 ± 6.8 years old. The most common glaucoma type and severity for the first and second eye were mild primary open angle glaucoma.

First and second eye phaco-MIGS cohorts were similar in preoperative medications, preoperative IOP, and target IOP. The first and second eyes were treated with a mean of 1.40 ± 1.3 and 1.26 ± 1.3 medications preoperatively. The preoperative mean IOP in the first and second eyes were 17.1 ± 4.8 mm Hg and 16.1 ± 4.3 mm Hg, respectively. The mean and median time in days between the first phaco-MIGS to the second phaco-MIGS procedure was 36 and 32 days, with a range of 10–137 days.

3.2. Change in intraocular pressure

Mean IOP and changes in IOP for the first and second phaco-MIGS eyes across postoperative visits are shown in Figure 1A and B. Between baseline and 6 months following surgery in the first phaco-MIGS eye, IOP decreased by 2.9 ± 1.3 mm Hg to 14.0 ± 3.2 mm Hg (20.9% reduction). For the second phaco-MIGS eye, IOP decreased by 1.9 ± 1.1 mm Hg to 14.0 ± 3.5 mm Hg (13.4 % reduction). The mean change in IOP for the first eye was larger across all postoperative time points compared to the second eye. However, there were no significant differences between mean change in IOP between groups at any time point.

Change in IOP between the first eye at postoperative visits >1 week demonstrated a moderately positive correlation with change in IOP for the second eye at 6 months, as shown in Figure 2A. The strongest correlation to second eye IOP change at 6 months was first eye IOP change at 6 months (R = 0.48; r^2 = 0.22; P < .001). Correlation between first and second eye change in IOP at the same postoperative time points were also evaluated. In addition to 6 month IOP change in fellow eyes, first eye change in IOP was shown to be moderately correlated with second phaco-MIGS eye IOP at 3 months following both surgeries (R = 0.42; r^2 = 0.18; P < .01) as shown in Figure 2B. Postoperative visits of 1 week and 1 month did not show significant correlations between fellow eyes.

An analysis of the number of IOP spikes across postoperative visits was performed. We defined IOP elevations as IOP >30 mm Hg or > 10 mm Hg above preoperative IOP. Our results showed that 9 events of IOP spikes occurred in the first eye at 1 week and 2 events of IOP spikes at 1 month with no additional spikes through 6 months. In the second eye, there were 7 events of IOP spikes at 1 week, 4 events at 1 month, and no further documented IOP spikes through 6 months. Using tetrachoric correlation, IOP spikes were shown to be weakly correlated between fellow eyes at 1 week and 1 month ([1 week] R = 0.35; [1 month] R = 0.70). The weak correlation at week 1 may be a result of the surgeon anticipating an IOP spike in the second eye following an IOP spike in the first eye and preemptively treating with medications.

Table 1

Demographic characteristics.

Parameters	156 eyes of 78 patients
Age* (years), mean ± SD	73.1 ± 6.8
Sex (no. (%))	
Female	43 (55.1)
Race/ethnicity (no. (%))	
African American	19 (24.4)
Asian	5 (6.4)
White	48 (61.5)
Other	6 (7.7)

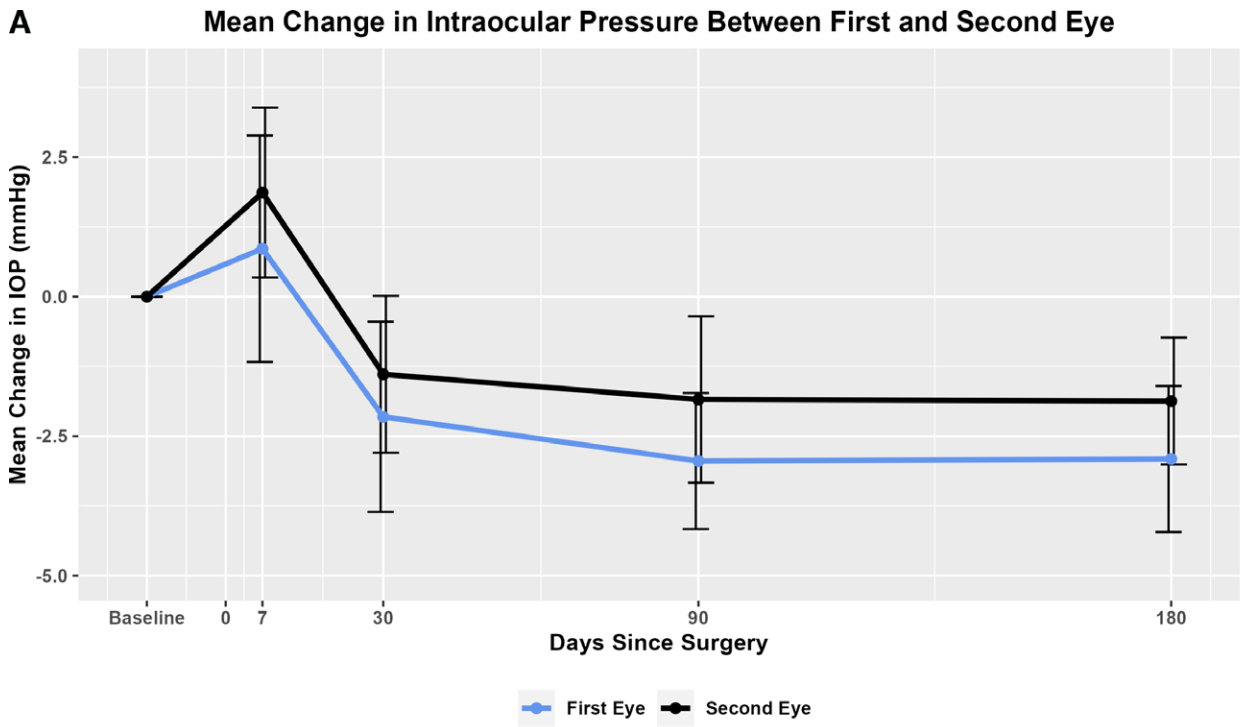
*Age based on years old at first phaco-MIGS procedure.

Table 2

Baseline ocular characteristics of first and second eyes.

Parameters	First eye phaco-MIGS N = 78	Second eye phaco-MIGS N = 78
Ocular characteristics		
Preoperative medications, mean ± SD	1.40 ± 1.3	1.26 ± 1.3
Preoperative IOP (mm Hg), mean ± SD	17.1 ± 4.8	16.1 ± 4.3
Target IOP	17.5 ± 3.3	18.1 ± 3.2
Vertical C:D ratio	0.69 ± 0.14	0.64 ± 0.15
VF MD (dB)	−6.57 ± 7.5	−4.00 ± 5.9
Postoperative follow-up visits	5.13 ± 1.1	3.41 ± 0.83
Prior glaucoma procedures (no. [%])		
None	67 (85.9)	65 (83.3)
Trabeculectomy	1 (1.3)	4 (5.1)
ALT	2 (2.6)	1 (1.3)
SLT	8 (10.3)	7 (9.0)
Yag PI	5 (6.4)	4 (5.1)
Glaucoma severity (no. [%])		
Mild	46 (59.0)	60 (76.9)
Moderate	18 (23.1)	9 (11.5)
Severe	14 (17.9)	9 (1.51)
Glaucoma type (no. [%])		
Primary open angle glaucoma	38 (48.5)	35 (44.9)
Primary angle closure glaucoma	25 (32.1)	22 (28.2)
Other	15 (19.2)	21 (26.9)
Surgery type (no. [%])		
Hydrus	27 (34.6)	27 (34.6)
OMNI surgical system	17 (21.8)	17 (21.8)
Combined OMNI + hydrus	16 (20.5)	16 (20.5)
Bent Ab interno needle goniotomy	13 (16.7)	13 (16.7)
Kahook dual blade	3 (3.8)	3 (3.8)
iStent	2 (2.6)	2 (2.6)

ALT = argon laser trabeculoplasty, C:D = cup-to-disc, IOP = intraocular pressure, SD = standard deviation, SLT = selective laser trabeculoplasty, VF MD = visual field mean deviation, Yag PI = YAG laser iridotomy.



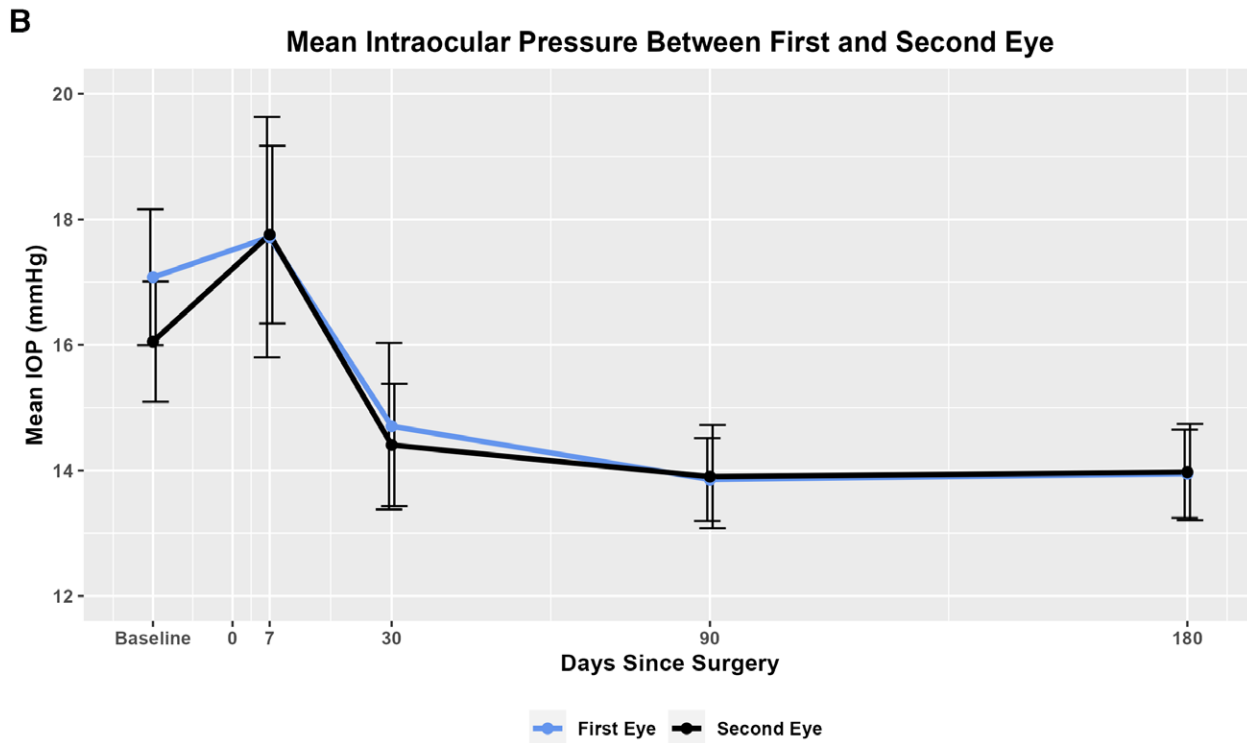


Figure 1. (A) Mean change in intraocular pressure between first and second eye. (B) Mean intraocular pressure between first and second eye.

3.3. Change in medications

Mean ocular hypotensive glaucoma medications and change in medications for the first and second phaco-MIGS eye across postoperative visits are shown in Figure 3A and B. From baseline to 6 months following surgery, first and second phaco-MIGS glaucoma medications decreased by 1.1 ± 0.3 and 1.0 ± 0.2, respectively. Mean glaucoma medications at 6 months were 0.3 ± 0.6 for both the first and second eye, respectively. The mean change in glaucoma medications between the first and second eye was not clinically significant. However, medication change was slightly greater for the first eye than the second eye at all postoperative time points >1 week.

Correlations between first eye changes in glaucoma medications at all postoperative time points and second eye medication change at 6 months were all significantly correlated (Fig. 4A). However, correlations between first and second phaco-MIGS eyes changes in glaucoma medications at all identical postoperative visit time points demonstrated a stronger correlation shown in Figure 4B. Overall, glaucoma medication change

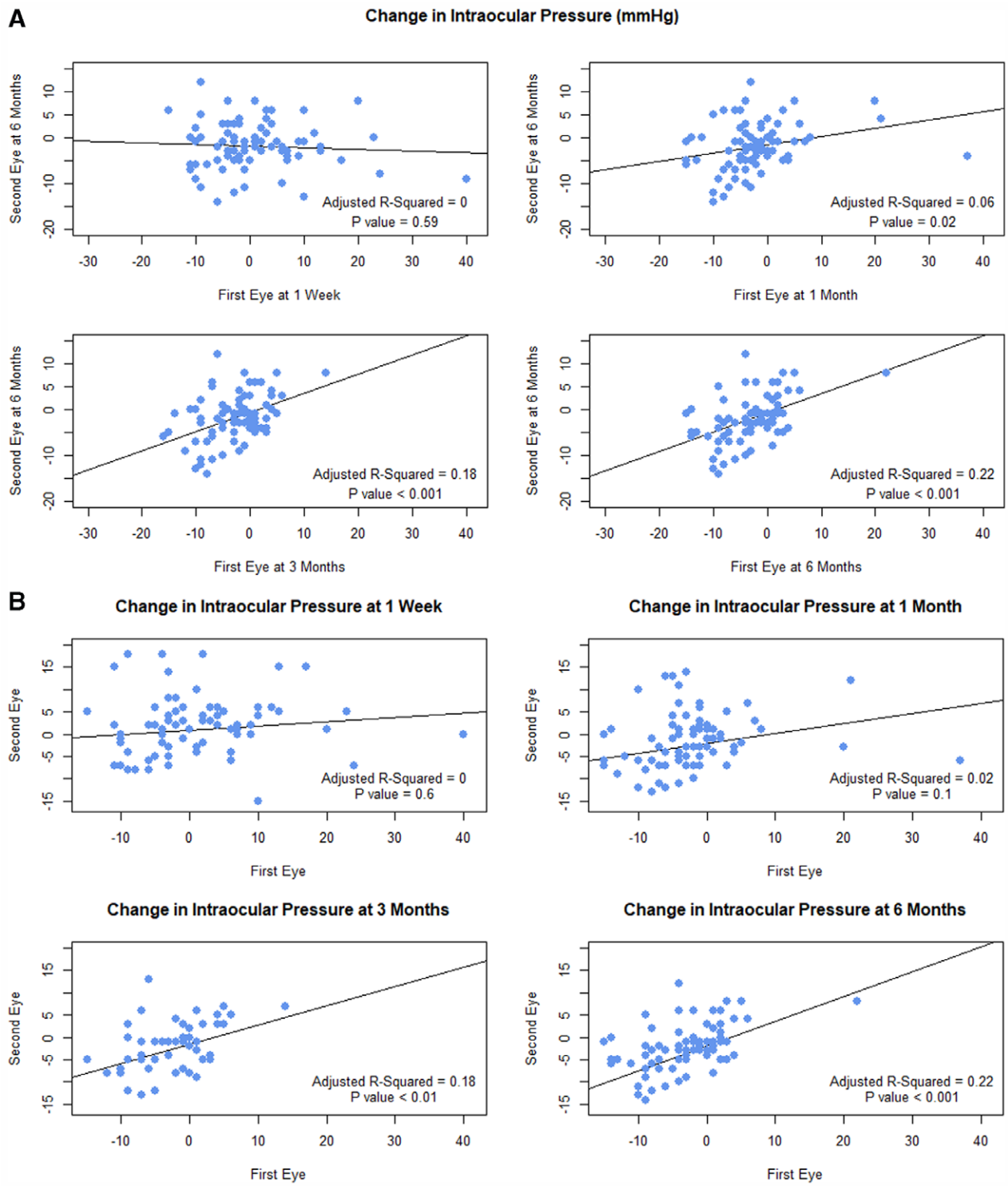


Figure 2. (A) Correlation between change in intraocular pressure of the first eye at 6 months and the fellow eye. (B) Correlation between change in intraocular pressure of fellow eyes.

demonstrated the strongest correlation after 6 months postoperatively ($R = 0.80$; $r^2 = 0.64$, $P < .001$).

3.4. Likelihood of success

Of first and second phaco-MIGS eyes, 82% and 83% of each cohort achieved success at 6 months, respectively. A stratified analysis of the number of first-eye phaco-MIGS patients categorized by the time in days until second-eye phaco-MIGS surgery is shown in Figure S1, Supplemental Digital Content, http://links.lww.com/MD/K690. A majority of second eye phaco-MIGS procedures occurred within 2 months of first phaco-MIGS eye surgeries, demonstrating clinicians' made decisions to proceed with second phaco-MIGS eyes procedures without knowledge of month 3 or 6 first eye outcomes.

McNemar tests between first eye surgical success at postoperative visits of 1 week, 1 month, 3 months, and 6 months and second eye surgical success at 6 months demonstrated

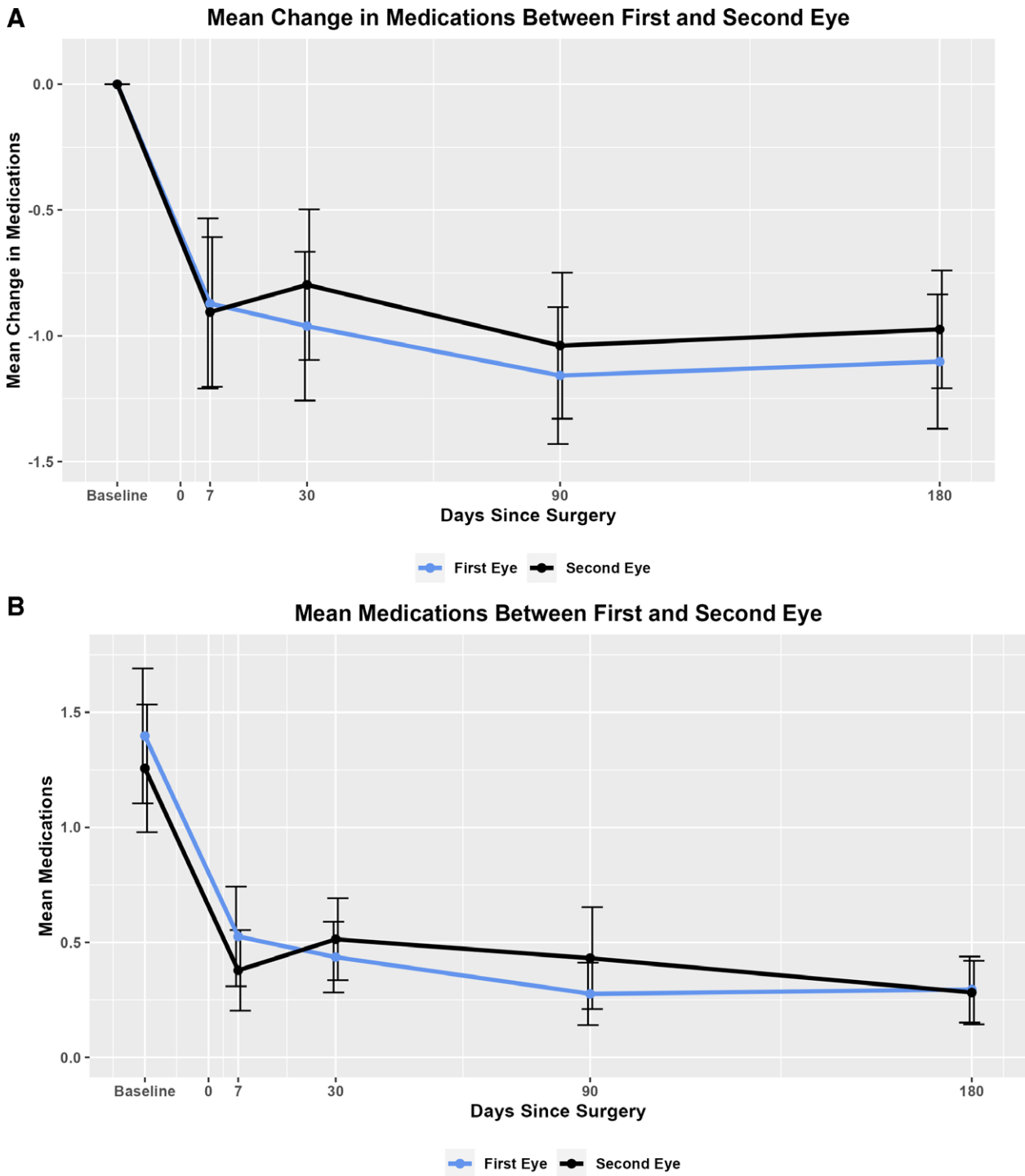


Figure 3. (A) Mean change in ocular hypotensive medications between first and second eye. (B) mean ocular hypotensive medications between first and second eye.

differences in the proportion of successes across postoperative time points (Table 3). There were statistically significant differences between the proportion of successes in the first eyes at 1 week and 1 month compared to the proportion of successes in the second phaco-MIGS eyes at 6 months. Between first eye success at 3 months and 6 months and second eye success at 6 months, the proportion of successful surgeries was shown to be similar. Univariate analysis showed that first-eye surgical success at month 3 and month 6 was significantly associated with greater surgical success rates in the second eye at 6 months (Table 4). First eye success at 3 and 6 months increased the likelihood of successful second eye surgery by 5-fold (OR, 5.14; 95% confidence interval [CI], 1.38–19.32; $P = .01$) and 27-fold (OR, 27.00; 95% CI, 6.58–135.53; $P < .001$), respectively. In contrast, the number of glaucoma medications and moderate to severe glaucoma staging of the first eye were associated with a significantly decreased likelihood of surgical success in the second eye. For every additional glaucoma medication taken, patients were half as likely to be successful at 6 months (OR, 0.57; 95% CI, 0.35–0.90; $P = .02$). Additionally, patients with severe or moderate glaucoma staging were 80% and 86% less likely than those with mild stage glaucoma to be successful ([moderate] OR, 0.18; 95% CI, 0.03–0.84; [severe] OR, 0.13; 95% CI, 0.02–0.60).

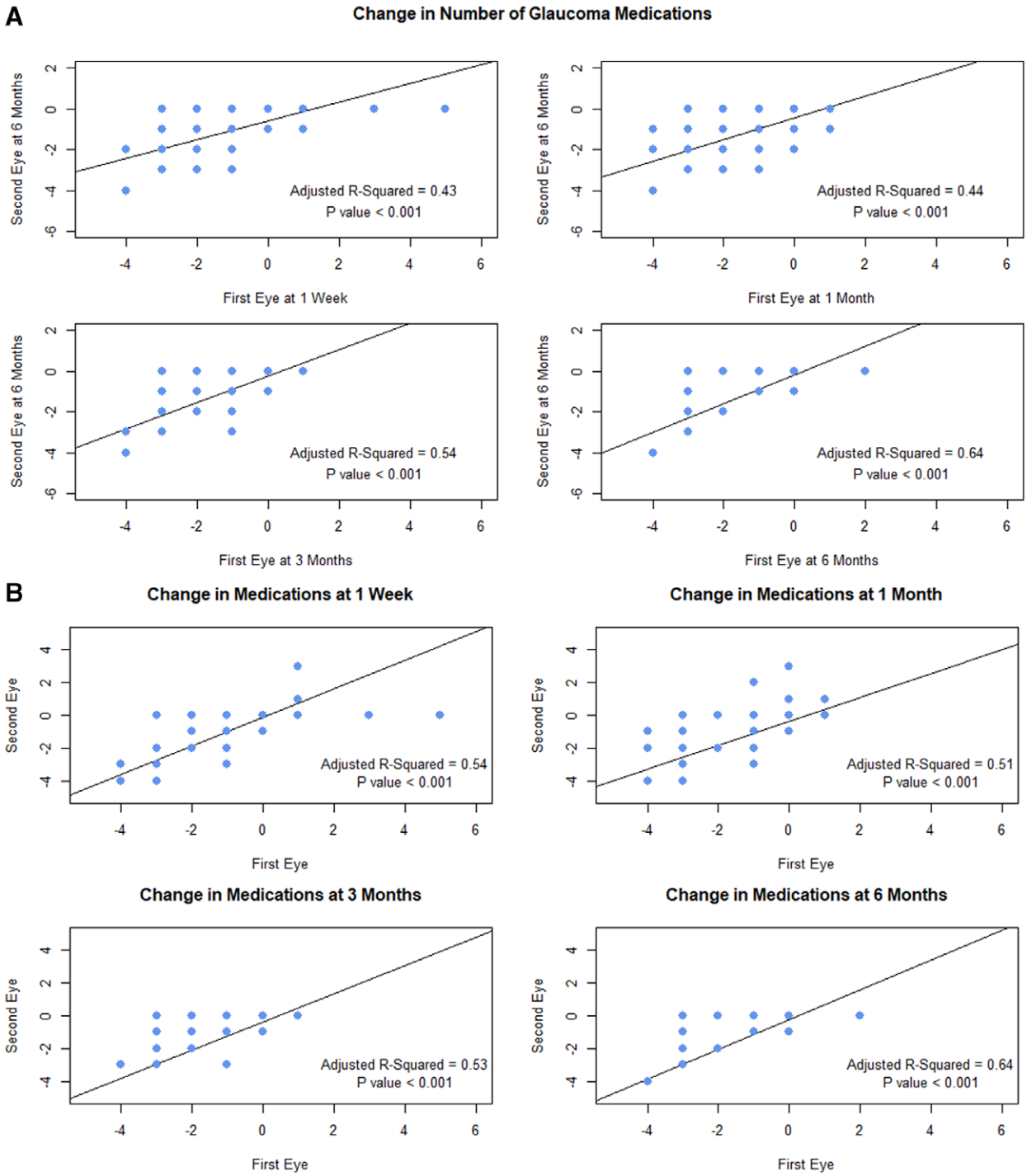


Figure 4. (A) Correlation between change in glaucoma medications of the first eye at 6 months and the fellow eye. (B) Correlation between change in glaucoma medications of fellow eyes.

A multivariate analysis of months 3 and 6 first eye success, sex, glaucoma severity, and the number of glaucoma medications showed first eye successful surgery at 6 months to be the sole predictor of second phaco-MIGS eye success (Table 5). Patients with successful first eye phaco-MIGS at 6 months were 20-fold more likely to have a successful second eye phaco-MIGS surgery at 6 months (OR, 20.67; 95% CI, 2.96–234.16; $P < .001$).

Kaplan–Meier survival analysis was used to assess surgical success rates between the fellow eyes (Fig. 5A). There were no significant differences in surgical success rates between the 2 cohorts ($P = .3$). Surgical success for the first eye was 55.1%, and for the second eye was 61.7%. A second Kaplan–Meier survival curve was created for second eye success where the second eyes were grouped based on whether their fellow first eye was a success or failure at 6 months. Of the second eyes, where the first eye was a failure, their success rate was 25.7%. In contrast, the success rate was 70.3% for second eyes, where the first eye was successful, as shown in Figure 5B ($P = .002$).

4. Discussion

With the increasing use of MIGS in glaucoma clinical management strategies, the utilization of predictive measures for

determining optimal candidates for MIGS surgery has potentially broad implications. Recent studies have evaluated various aspects of patient and demographic characteristics associated with surgical success outcomes of different types of MIGS procedures.[5,6,18] To the best of our knowledge, this is the first study to evaluate first-eye outcomes at various postoperative time points as predictor factors of success in second-eye phaco-MIGS surgeries. We also analyzed the correlations in IOP and glaucoma medication change between fellow eyes as a predictive measure for second-eye IOP and glaucoma medication responses at 6 months postoperatively. Our results demonstrate that changes in IOP and glaucoma medications were modestly predictive of IOP and glaucoma medication changes between study and fellow eyes at 6 months. Additionally, patients with successful first eye surgeries at 6 months were more likely to have successful second eye surgeries at 6 months. These findings highlight the real-world implications of utilizing 6-month outcomes in first eyes as a promising predictive tool for patients who require bilateral surgery.

Table 3

Likelihood of 6 month second-eye success based on first-eye success at various postoperative timepoints.

Parameters		Second eye month 6		
First eye		Failure	Success	*P*
Week 1	Failure	7	35	≤.001***
	Success	6	30	
Month 1	Failure	6	23	.006**
	Success	7	42	
Month 3	Failure	6	9	.80
	Success	7	54	
Month 6	Failure	9	5	1
	Success	4	60	

*$P \leq .05$.
**$P \leq .01$.
***$P \leq .001$.

4.1. Correlations in intraocular pressure change

Our results demonstrated that change in IOP for the first phaco-MIGS eye at 3 months and 6 months is positively correlated to IOP change in second phaco-MIGS eye at 6 months. While these correlations are weak to moderately strong, these results suggest that first-eye responses to changes in IOP at and after 3 months postoperatively may be used to predict changes in IOP for the second eye following phaco-MIGS surgeries.

Some studies have suggested that glaucoma surgery in one eye may cause IOP elevations in the untreated eye up to 6 months following IOP-lowering procedures.[19–21] Meanwhile, other studies have reported no significant changes in IOP of the fellow eye after glaucoma surgery.[22] In our study, the baseline IOP for the second eye phaco-MIGS was, on average, lower than in the first eye cohort, with a mean of 16.1 mm Hg compared to 17.1 mm Hg. However, we recognize these differences in baseline IOP may be attributed to surgeons' preferred practice pattern to operate first on the eye with higher IOP. Despite these slight differences in baseline IOP, the majority of both first and second eyes were staged with mild glaucoma. Interestingly, the mean IOP for both the first and second eye at 6 months was 13.9 mm Hg. These findings suggest that similar changes in IOP are most likely attributed to the individual's similar constitutional response and recovery from each MIGS or regression to the individual's mean IOP.

A potential explanation for the absence of correlation in IOP change between the first eye at postoperative time points of 1 week and 1 month compared to the second eye may be a consequence of corticosteroid use and variability in the extent to which eyes demonstrate a steroid response. There were 11 events of IOP spikes in both the first and second eye within the first month of surgery, with no additional spikes in either eye through 6 months. These results suggest that interpretation

Table 4

Univariate analysis of first eye characteristics with successful second eye surgery.

Univariate analysis	Odds ratio	Confidence interval	*P*
Proportion of success following first eye surgery at postoperative visits (reference: first eye failure at each time point)			
1 week	1.00	0.30–3.42	1.00
1 month	1.57	0.46–5.27	.47
3 months	5.14	1.38–19.32	.01**
6 months	27.00	6.58–135.53	≤.001***
Demographic characteristics			
Age	1.00	0.91–1.09	.95
Sex (reference group: female)			
Male	0.30	0.07–1.01	.06
Race/ethnicity (reference group: Asian)			
White	1.25	0.06–10.00	.85
African American	1.33	0.06–14.24	.82
Other	1.25	0.04–39.01	.89
Ocular characteristics of first eye			
Glaucoma severity (reference group: mild severity)			
Moderate	0.18	0.03–0.84	.03*
Severe	0.13	0.02–0.60	.01**
Glaucoma type (reference group: primary angle closure glaucoma)			
Other glaucoma types	1.28	0.38–4.37	.69
Baseline IOP	1.07	0.94–1.24	.30
Number of baseline Glaucoma Medications	0.57	0.35–0.90	.02*
Prior glaucoma procedures	1.00	0.26–6.47	1.00
Treating surgeon	1.05	0.50–2.15	.90

IOP = intraocular pressure.
*$P \leq .05$.
**$P \leq .01$.
***$P \leq .001$.

Table 5

Multivariate analysis of first eye postoperative characteristics with successful second eye surgery.

Multivariate analysis	Odds ratio	Confidence interval	*P*
Proportion of success following first eye surgery at postoperative visits (reference: first eye failure at each time point)			
3 months	0.98	0.10–6.62	0.98
6 months	20.67	2.96–234.16	≤ 0.001***
Demographic and ocular characteristics of first eye			
Sex (reference group: Female)			
Male	0.27	0.04–1.44	0.14
Glaucoma severity (reference group: Mild)			
Moderate	0.72	0.09–6.33	0.75
Severity	0.93	0.11–8.92	0.95
Number of baseline Glaucoma Medications	0.70	0.37–1.30	0.25

*$P \leq .05$.
**$P \leq .01$.
***$P \leq .001$.

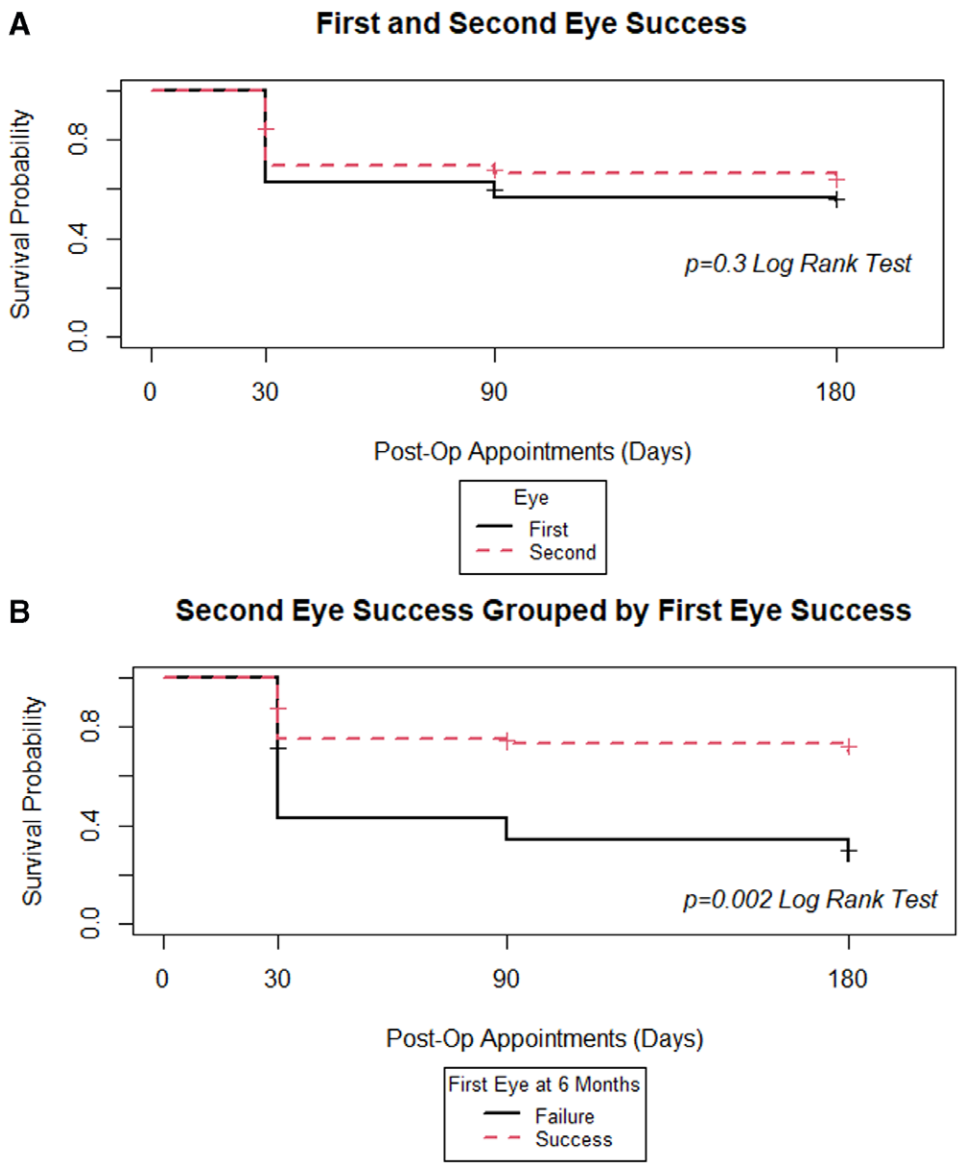


Figure 5. (A) Kaplan–Meier survival curve for first and second eye success. B) Kaplan–Meier survival curve for second eye success, where second eyes were grouped based on whether their fellow first eye was a success or failure at 6 months.

of IOP change before 1 month for predicting responses of IOP change in fellow eyes may be unreliable due to the influence of early postoperative corticosteroid use. The real-world implications of these findings suggest that a surgeon may elect to proceed with the second eye if the patient has mild glaucoma and minimal medications. However, in more severe cases,

a surgeon might consider waiting on first eye outcomes before electing to proceed with phaco-MIGS in the second eye.

4.2. Correlations in glaucoma medications change

Change in glaucoma medications was significant for the first eye at all postoperative time points through 6 months compared to the second eye at 6 months. While these relationships were strongly correlated, comparison of the change in glaucoma medications between the study and fellow eyes at the same time points were slightly more correlated as expected. A potential interpretation of these results for glaucoma medication change may be attributed to similar responses in second eyes compared to first eyes that necessitate similar glaucoma medication regimens. Ancillary to this hypothesis, these trends may be a reflection of clinicians' underlying practice patterns for introducing or discontinuing glaucoma medications bilaterally. Overall, these findings highlight the capacity for surgeons to set expectations with high certainty for patients regarding the medication decrease they are likely to achieve for their second eye.

4.3. Likelihood of success

The results of this study support utilizing information on surgical outcomes at 6 months in first eyes following phacoemulsification and MIGS as a predictive measure for outcomes in fellow eyes in patients requiring bilateral surgery. We hypothesize that the strong association between the eyes at 6 months may be attributed to intrinsic physiologic responses from the same individuals with similar underlying pathologies. Moreover, the time point of 6 months may be most predictive of success compared to earlier postoperative time points because of the duration of time given for the eye to achieve stability and recovery following postoperative inflammation and corticosteroid use. As previously mentioned, all IOP spikes occurred in the first month following surgery in the first eye.

Overall, the results of this study provide evidence for the continued use of MIGS as a valuable tool for glaucoma management. Thus, the high success rate of MIGS shown in this study and many other previously reported studies introduces the dilemma of whether the benefits of waiting 6 months following first eye surgery outweigh the potential cost of glaucomatous damage progression in the fellow eye. In addition, refractive concerns such as living with prolonged anisometropia, purchasing multiple pairs of glasses, or changing the first eye's refractive target due to the long wait must be considered. Many clinicians with patients requiring bilateral IOP-lowering surgical therapy may elect to proceed with second eye surgery without knowing the outcomes of the first eye, given the overall high probability of MIGS success. Yet, this 6 month waiting period following first eye outcomes may be applicable for patients who cannot afford to have a failed outcome for their second eye. Additionally, this 6 month waiting period may apply to patients where financial consciousness plays a role in treatment management decisions. A recent study found that traditional glaucoma filtering surgeries and selective laser trabeculoplasty are more cost-effective than MIGS at reducing IOP.[23] Therefore, it may be more economical for these patients to delay second eye surgery to determine the true IOP and glaucoma medication lower benefits of MIGS.

4.4. Limitations

Limitations include the modest sample size, the limited follow-up length of 6 months, and the retrospective study design. Additionally, systemic factors such as diabetes, preexisting ocular conditions, and ocular comorbidities other than glaucoma were not evaluated. Because our study period overlaid the COVID-19 era, we recognize that the mean time in days between first and second eye MIGS surgeries may be biased by individuals' choices to undergo or temporarily postpone elective surgery. Our study also included several types of MIGS procedures; however, we acknowledge that recent studies have demonstrated superior rates of IOP and medication-lowering effects with some MIGS surgeries compared to others, as with the COMPARE study evaluating outcomes of Hydrus and 2-iStents.[24] Additionally, we understand that a clinician's practice management (e.g., a surgeon's familiarity with a particular MIGS type or a surgeon's decision to modify postoperative glaucoma medications regimens) may have influenced our study's outcomes. However, a univariate analysis found that surgeon was not significantly associated with the likelihood of second eye phaco-MIGS surgical success. Lastly, we recognize that our data is limited to patients who underwent phaco-MIGS in both eyes; therefore, we were unable to account for potential cases where first-eye MIGS demonstrated a poor outcome and second-eye surgery was abandoned. However, a stratified analysis of the number of first-eye phaco-MIGS patients categorized by the time in days until second-eye phaco-MIGS surgery in Figure S1, Supplemental Digital Content, http://links.lww.com/MD/K690 demonstrated that clinicians were proceeding with second-eye surgeries prior to knowing longitudinal outcomes of first phaco-MIGS eyes. Randomized controlled trials with longer durations of follow-up periods are warranted to validate and quantify the degree of predictive capacity of first eye MIGS outcomes to estimate second-eye MIGS outcomes.

4.5. Conclusion

Postoperative changes in IOP and glaucoma medications at 6 months in the first phaco-MIGS eye were modestly predictive of IOP and glaucoma medication changes in fellow eyes at 6 months. Patients who were successful at 6 months for first eye phaco-MIGS procedures were also more likely to have a successful second eye phaco-MIGS procedure at 6 months. These findings support the use of 6-month outcomes in first eyes as a promising predictive tool for patients requiring bilateral surgery when the clinical scenario affords that length of delay between the first and second eye. Further prospective studies should be conducted on whether outcomes following the first phaco-MIGS eyes can be utilized as a predictive measure for clinical outcomes in a patient's fellow eye.

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